Statistical questions

1 State whether the following questions are biased or fair:

a Do you want a soup for lunch or a sandwich?

b What do you do on a Sunday morning?

c Do you like these shoes?

d Do you prefer this shirt or the one on the shelves at the moment?

e Do you think the government should be allowed to cut down some of the oldest trees in the area to construct a metro railway line in the city?

f Do you prefer newspapers or news on television?

g Do you prefer the full time degree program or part time degree program?

h Should the government enforce a minimum drinking age for its citizens?

i Do you eat at least the recommended number of servings of fruits and vegetables to ensure a healthy and long life?

j Do you think bike helmets should be mandatory for all bike riders?

k Do you prefer the natural beauty of hardwood floors in your home?

l Do you exercize regularly?

m Do you feel that the TV news is an inaccurate portrayal of life's problems?

n Don't you think this newspaper is biased?

o Do you prefer the look and feel of thick lush carpeting in your living room?

p Do you take these extra strength multi-vitamins to supplement your diet?

2 State whether each of the following biased questions are leading or emotive:

a Do you want a nutritious risotto for lunch or the usual sandwich?

b Do you watch TV on a Sunday morning like everyone else?

c I don't like these shirts. Do you?

d Do you prefer this red shirt or the ordinary one on the shelves at the moment?

3 Adults attending a local cinema were asked the question: "How many times did you see a movie at this cinema last year?"
Explain why this question is may give inaccurate results.

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Sampling

4 Explain why the following samples are biased:

a Hannah is surveying customers at a shopping precinct. She wants to know which stores customers shop at the most. She walks around an entertainment store and chooses 30 customers from the store for the survey.

b A TV station wants to know what the most popular type of music is, so they ask listeners to contact them and vote for their favorite type of music.

5 The community health nurse wants to survey the students in a school about their eating habits. At lunchtime, she stands by a vending machine and surveys every student who purchases something from the machine.
Explain why this sample is biased.

6 State whether the following scenarios use biased sampling methods:

a A community nurse wants to know the average height of all 7th graders that attend the school where she visits, so she measures the height of all the basketball players.

b A city councilman asks members of the ice hockey team if they would prefer a new skateboard park or a new ice-skating rink to be built as the new building project.

c The lifeguard of a water park wants to determine which water rides are enjoyed the most so he asks every tenth person who leaves the park to list their three favorite rides.

7 A school principal wants to estimate the number of students who ride a bicycle to school. State whether the following samples would avoid bias:

a All students who are in the school band.

b 8 students in the hallway.

c Ten students from each grade, chosen at random.

d 130 students during the lunch periods.

8 Four lucky people from a group of 215 each stand to win a tablet. Every contestant is given a different number between 1 and 215, and the winners are selected by generating a random number uniformly between 0 and 1. To ensure there is an equal chance of each contestant winning, the number is multiplied by 215 and then rounded up.
In this case, the numbers generated were:

$$0.152,\ 0.534,\ 0.352,\ 0.795$$

Convert the generated numbers into the numbers of the four winning contestants.

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9 Beth is interested in which students from her school catch public transport.
Determine whether the following sampling methods are likely to be biased or not:

a Selecting every 10th person on the bus she catches.

b Selecting every 10th person on the student list.

c Selecting the first 50 students that arrive in the morning.

d Selecting by having a computer randomly choose student numbers.

10 Georgia wants to know how the people in America are going to vote in an upcoming referendum. She selects 50 random people from her city to interview.
Explain why this sampling might give inaccurate results.

11 Justin wants to know how the students at his school might vote for the next class president. He leaves three flyers in each class to complete by whatever students want to complete them.
Explain why this sampling might give inaccurate results.

Sampling and measurement errors

12 Select the type of sampling error for each of the scenarios from the following three options:

- The sample is too small.
- The sample is not random.
- The sample does not adequately represent the population.

a Tina wants to know which new smart phone she should buy and decides to base her decision on other people's opinions. She decides to interview some people at the retirement home where her grandparents live.

b In order to gauge how Americans feel about immigration, a local newspaper sent a reporter to go door-knocking in a local suburb and to ask a series of questions.

c For the 2020 US Presidential election, a sample of $10\,000$ people in New York were polled to see who they would vote for.

d Vincent is conducting research on the average price of 1 pint of milk in Indianapolis. He visits two shopping centers, two delis and one convenience store to collect this information.

13 In 2016, the American census was conducted online for the first time and the website crashed. Is this an example of sampling error or measurement error or neither?

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14 An online store sent a survey to all people who had bought the most recent book in the Harry Potter series and asked them “Which do you prefer, the Harry Potter books or the Harry Potter films?”

a Explain why there is a sampling error in this sample.

b Explain why there is a measurement error in this question.

15 Describe the types of measurement errors for each of the following scenarios:

a A survey of a large random sample with participants from all over America was taken and the following question asked: "Do you think the government is doing enough to stop the increase in violent crime on our streets?"

b A survey of a large random sample with participants from all over America was taken and the following question asked: “Do you regularly recycle all your waste paper and plastic?”

16 In order to determine the most common non-English language spoken in Pittsburgh, adults and children at a football game were asked: “Other than English, what language do you speak at home?”

a Explain why this situation represents a sampling error.

b Explain why this question also represents a measurement error.

17 A survey to assess American parents' attitudes towards private schools was conducted. Parents with young children enrolled at Pebbles Daycare centers around America were surveyed and asked the following question: “Which is better? Private schools or public schools?”
Describe some possible sampling errors with this survey.

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